

# Text-Conditioned Semantic Aggregation for Audio-Based Multiple-Instance Learning in Depression Detection

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**Abstract.** Automatic depression detection from semi-structured interviews has gained increasing relevance as a scalable and non-invasive complement to conventional clinical assessment, particularly in digital and remote-care contexts. However, modeling long-form interviews remains challenging due to their weakly supervised nature: a single subject-level label must be inferred from multiple heterogeneous conversational segments. This setting naturally fits the Multiple Instance Learning (MIL) paradigm, where each interview is represented as a bag of instances associated with a global label. Nevertheless, existing MIL-based approaches typically rely on implicit attention or aggregation mechanisms that compress local evidence into a final decision without explicitly grounding segment importance in structured clinical knowledge. Here, we propose a text-conditioned attention-based MIL framework, termed TCAMIL, for audio-based depression detection that incorporates symptom-level semantic alignment prior to aggregation. A semantic relevance filtering stage is implemented through a cross-encoder architecture to match interview text segments with Patient Health Questionnaire-8 (PHQ-8) symptom items, preserving only clinically aligned instances. This filtering enables the construction of noise-reduced MIL bags composed of symptom-relevant audio fragments, reducing the impact of semantically diverse and weakly informative content. The selected instances are then processed using a domain-adapted encoder and aggregated via trainable attention pooling to produce subject-level predictions under weak supervision. Experiments conducted on the DAIC-WOZ dataset demonstrate improved performance. Additionally, qualitative analyses indicate enhanced interpretability, as the model assigns higher relevance to segments aligned with structured symptom descriptors.

**Keywords:** Multiple Instance Learning · Depression Detection · PHQ-8 Semantic Alignment

## 1 Introduction

Automatic depression detection has emerged as a critical complement to conventional clinical assessment, driven by the increasing demand for scalable, reproducible, and non-invasive screening tools in digital and remote-care environments (9). Although structured clinical interviews and standardized questionnaires remain central to mental health evaluation, their outcomes may still be influenced by clinician expertise, patient self-presentation, recall bias, and contextual variability across sessions (10). These factors motivate the development of computational systems capable of providing consistent and objective support for clinical decision-making. In recent years, advances in artificial intelligence have enabled the automated analysis of semi-structured interviews through models that process linguistic content and, in several cases, integrate acoustic and visual cues to enhance robustness and clinical validity (14). Such multimodal frameworks aim to capture subtle

behavioral and paralinguistic signals that may reflect affective and cognitive patterns associated with depressive symptomatology.

Within this evolving landscape, prior research has explored hierarchical recurrent architectures, multimodal deep-learning pipelines, transformer-based encoders, and large language models for detecting depressive markers in interview data (2). More recently, Multiple Instance Learning (MIL) formulations and interpretable pretrained representations have been introduced to better accommodate long interview transcripts, enabling models to identify or score clinically informative segments without requiring fine-grained annotations (15). Despite these advances, most approaches rely on trainable attention or aggregation mechanisms that compress distributed local evidence into a single global prediction. In many cases, the relative contribution of each segment is modeled implicitly, remaining entangled with the final bag-level decision and lacking explicit grounding in structured clinical knowledge (8). When an interview is represented as a sequence of interventions, it can be naturally formulated as a bag of instances associated with a single global label, where each utterance or response contributes unequally to the final decision (11). Under this MIL perspective, the learning objective becomes the identification of diagnostically meaningful fragments without access to explicit instance-level annotations (12). This problem is particularly challenging in long-form interviews, where conversational segments are semantically heterogeneous and their diagnostic value may fluctuate due to ambiguity, sparse emotional expression, topic shifts, or inconsistent self-disclosure (13). Consequently, a central open question is not only how to detect depression from extended interviews, but also how to systematically identify clinically relevant segments within a weakly supervised and potentially noisy decision framework (1).

To address this gap, we propose a text-conditioned attention-based MIL framework (TCAMIL) for audio-based depression detection. The proposed approach integrates structured symptom knowledge prior to aggregation through three key components. First, we introduce semantic relevance filtering using a cross-encoder architecture to match interview text segments with Patient Health Questionnaire-8 (PHQ-8) symptom items, preserving only clinically aligned instances for downstream modeling. Second, we construct noise-reduced MIL bags by restricting them to symptom-relevant audio fragments, thereby mitigating the adverse impact of semantically diverse and weakly informative conversational content. Third, we employ attention-based weakly supervised aggregation through a domain-adapted encoder and trainable attention pooling to learn bag-level depression predictions from filtered instances. Our framework is evaluated on the DAIWOZ database, demonstrating its effectiveness for clinically guided depression detection in long-form interviews by explicitly aligning interview evidence with structured symptom descriptors.

The agenda is as follows. Section 2 describes the materials and methods. Sections 3 and 4 present the experimental setup and results, respectively. Finally, Section 5 concludes the paper.

## 2 Materials and Methods

### 2.1 DAIC-WOZ Dataset

The proposed framework was evaluated on DAIC-WOZ, a widely used benchmark dataset for automatic depression assessment based on Wizard-of-Oz clinical interviews (4). As a subset of the Distress Analysis Interview Corpus, DAIC-WOZ provides multimodal recordings, transcripts, and depression-related annotations collected during interactions between participants and an embodied virtual interviewer (Ellie). In this study, we use the official subject-independent partition, consisting of 107 training sessions and 35 validation sessions, including both healthy-control and depressed

participants. The data exhibits a moderate class imbalance, with approximately 30% depressed and 70% non-depressed participants, and a gender distribution of roughly 57% female and 43% male subjects. Audio recordings were acquired at 16 kHz and distributed in scrubbed form to preserve privacy. Since the participant channel may still contain residual interviewer speech and masked segments, transcript timestamps are used to perform speaker-aware temporal cropping, retaining only participant-response intervals.

In addition to bag-level labels, DAIC-WOZ provides the official PHQ-8 metadata for each session, including the binary depression label, the total PHQ-8 score, and the item-wise responses. These annotations are directly integrated into our framework as symptom-level clinical anchors for semantic instance selection. Specifically, the PHQ-8 covers eight core symptom dimensions: loss of interest (PHQ8\_NoInterest), depressed mood (PHQ8\_Depressed), sleep disturbance (PHQ8\_Sleep), fatigue (PHQ8\_Tired), appetite alteration (PHQ8\_Appetite), feelings of failure or guilt (PHQ8\_Failure), concentration difficulties (PHQ8\_Concentrating), and psychomotor changes (PHQ8\_Moving). Following the dataset protocol, PHQ-8 responses are encoded on a 0–3 ordinal scale, yielding a total score in the range  $[0, 24]$ , and a score  $\geq 10$  is used to define clinically relevant depressive symptomatology. Because this language-guided filtering depends on interviewer prompts, sessions with missing or severely incomplete interviewer transcripts are excluded.

## 2.2 Text-Conditioned Attention-Based Multi-Instance Learning (TCAMIL)

Let  $q_{ij}$  denote the interviewer question associated with the  $j$ -th temporal segment of subject  $i$ , and let  $\mathcal{P} = \{p_k\}_{k=1}^8$  be the PHQ-8 item texts (symptom descriptors). A Transformer-based cross-encoder  $\check{f}_{\text{CE}}$  jointly encodes each pair  $(q_{ij}, p_k)$  and produces a contextual semantic score:  $\check{r}_{ij}^{(k)} = \check{f}_{\text{CE}}(q_{ij}, p_k) \in \mathbb{R}$ . Then, for each subject  $i$ , we retain a clinically aligned subset of segments by selecting, for each PHQ item  $k$ , the most semantically compatible segment (or top- $S$  segments). Using the unweighted scores, a representative index is computed as:

$$j_i^{(k)} = \arg \max_{j \in \{1, \dots, m_i\}} \check{r}_{ij}^{(k)}. \quad (1)$$

The resulting text-guided, fixed-cardinality bag of selected instances is:  $\tilde{B}_i = \{\mathbf{x}_{i, j_i^{(k)}}\}_{k=1}^8$ , where  $\mathbf{x}_{i, j_i^{(k)}}$  denotes the acoustic segment of size  $T_{j_i^{(k)}}$  extracted from the participant-response audio aligned to question  $q_{ij}$ . This selection stage acts as a clinically grounded denoising operator: it suppresses semantically generic turns and preserves symptom-relevant evidence prior to weakly supervised MIL training for audio-based depression detection.

Next, given  $N$  subjects, we consider pairs  $\{\tilde{B}_i, y_i\}_{i=1}^N$ , where  $y_i \in \{0, 1\}$  denotes the subject-level depression label and  $\tilde{B}_i$  is the PHQ-aligned bag produced by the semantic matching stage. Each retained audio segment is represented by a fixed-dimensional acoustic descriptor  $\mathbf{x}_{ij} \in \mathbb{R}^d$  (eGeMAPS in our experiments (3)), and the bag is written as  $\tilde{B}_i = \{\mathbf{x}_{ij}\}_{j=1}^{\tilde{m}_i}$ , where  $\tilde{m}_i = 8$  (one segment per PHQ item) or  $\tilde{m}_i = 8S$  under top- $S$  selection. The objective is to learn a bag-level predictor  $\hat{y}_i = \mathcal{F}(\tilde{B}_i) \in [0, 1]$  under weak supervision. Following attention-based deep MIL (5), the mapping  $\mathcal{F}$  is decomposed into three parametric components:

$$\hat{y}_i = (f_{\text{cls}} \circ f_{\text{agg}} \circ f_{\text{enc}})(\tilde{B}_i). \quad (2)$$

The instance encoder  $f_{\text{enc}} : \mathbb{R}^d \rightarrow \mathbb{R}^M$  is implemented as a multilayer perceptron (MLP) that maps each acoustic descriptor into a latent embedding:  $\mathbf{h}_{ij} = f_{\text{enc}}(\mathbf{x}_{ij}) \in \mathbb{R}^M$ . Thus, each bag is transformed into a set of embeddings  $\mathbf{H}_i = \{\mathbf{h}_{ij}\}_{j=1}^{\tilde{m}_i}$ .

The aggregation module  $f_{\text{agg}}$  computes attention weights to quantify the contribution of each semantically selected segment to the bag-level decision. Instance embeddings are first projected into a latent attention space:  $\mathbf{u}_{ij} = \tanh(\mathbf{V}\mathbf{h}_{ij})$ , where  $\mathbf{V} \in \mathbb{R}^{L \times M}$  is trainable. Normalized attention coefficients are then computed as:

$$a_{ij} = \frac{\exp(\mathbf{w}^\top \mathbf{u}_{ij})}{\sum_{k=1}^{\tilde{m}_i} \exp(\mathbf{w}^\top \mathbf{u}_{ik})}, \quad (3)$$

with learnable vector  $\mathbf{w} \in \mathbb{R}^L$ . Afterward, the bag embedding is obtained as:

$$\mathbf{z}_i = \sum_{j=1}^{\tilde{m}_i} a_{ij} \mathbf{h}_{ij}, \quad (4)$$

which defines the output of  $f_{\text{agg}}$ .

The classifier  $f_{\text{cls}} : \mathbb{R}^M \rightarrow [0, 1]$  maps the aggregated representation to a depression probability:  $\hat{y}_i = \sigma(\tilde{\mathbf{W}}\mathbf{z}_i + \tilde{\mathbf{b}})$ , where  $\tilde{\mathbf{W}} \in \mathbb{R}^{1 \times M}$ ,  $\tilde{\mathbf{b}} \in \mathbb{R}$ , and  $\sigma(\cdot)$  denotes the sigmoid function.

Lastly, training is performed using Binary Focal Loss to address class imbalance and emphasize difficult cases, yielding:

$$\mathcal{L}_{\text{FL}} = -\frac{1}{N} \sum_{i=1}^N [\alpha y_i (1 - \hat{y}_i)^\gamma \log(\hat{y}_i) + (1 - \alpha) (1 - y_i) \hat{y}_i^\gamma \log(1 - \hat{y}_i)], \quad (5)$$

where  $\gamma \geq 0$  is the focusing parameter and  $\alpha \in [0, 1]$  balances class contributions.

The resulting framework, termed TCAMIL, is depicted in Figure 1.

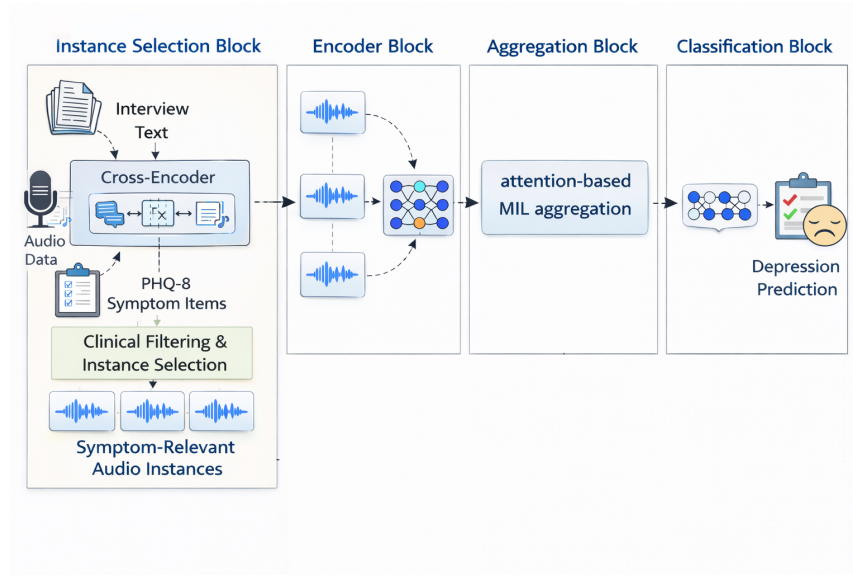


Fig. 1: Text-conditioned attention-based MIL framework (TCAMIL) for depression detection.

### 3 Experimental Setup

Experiments were conducted on the DAIC-WOZ benchmark for automatic depression assessment using the subject-independent partition described in Section 2.1. From the 142 subjects available in DAIC-WOZ, 140 sessions were effectively retained for the semantic selection stage, since two subjects lacked interviewer transcripts. In our experiments, each retained audio segment was represented by the official eGeMAPS descriptor with dimensionality  $d = 88$ , extracted with the standard openSMILE toolkit and the official eGeMAPS configuration (3). In parallel, the PHQ-8 item descriptions were used as structured semantic anchors to compute the semantic relevance scores, which were used to retain a single interview question associated with each PHQ-8 item before MIL aggregation, e.g.,  $S = 1$  (see Section 2.2).

We compared the proposed TCAMIL framework against three non-MIL baselines: SVM, Random Forest, and FAD3QN (6), together with one MIL baseline based on gated-attention Deep MIL (5). Semantic matching was estimated using an MPNet-based sentence transformer and a MiniLM-based cross-encoder, with a cosine-softmax temperature of 1.0. In the MIL model, the instance encoder employed an embedding dimension of 500, while the attention module operated in a latent space of dimension 64. For the SVM and Random Forest, subject-level representations were constructed exclusively from the participant’s speech segments. Specifically, eGeMAPS features were extracted from patient-response audio only (excluding interviewer speech), and fixed-length representations were obtained by concatenating descriptors computed over ten randomly selected segments (7). Importantly, FAD3QN does not perform semantic segment selection or attention-based aggregation. Instead, it randomly selects ten audio segments per subject, extracts eGeMAPS features from each segment, and concatenates them into a fixed-length representation. This aggregated feature vector is then treated as a single input instance and processed through a conventional deep learning architecture.

All MIL-based neural models were trained for 200 epochs using AdamW with learning rate  $3 \times 10^{-4}$  and weight decay  $10^{-4}$ . To address class imbalance, Binary Focal Loss with  $\gamma = 2$  was used. Model selection was based on the checkpoint with the highest validation accuracy. To obtain an unbiased validation estimate, we used a stratified train-validation split with fixed seed 0 and shuffling enabled, while final testing was performed on the official test partition provided by the dataset. All experiments were executed in a Kaggle environment with two NVIDIA Tesla T4 GPUs (T4 $\times$ 2), using Python 3.10.2, NumPy 2.0.2, PyTorch 2.9.0+cu126, and Scikit-Learn 1.6.1. Performance was assessed using standard binary-classification metrics, namely Accuracy, Precision, Recall, F1-score, and ROC-AUC. Code is available at GitHub repository

### 4 Results and Discussion

We assess whether the cross-encoder can reliably map interview prompts onto the PHQ-8 symptom space and whether this mapping yields clinically meaningful audio instance selection. Figure 2a shows the semantic affinity landscape between interviewer prompts (rows) and PHQ-8 items (columns) for an illustrative participant. The matrix exhibits a high-contrast and sparse structure: most generic prompts produce uniformly low scores, whereas a small subset of clinically targeted questions yields sharp maxima concentrated on specific PHQ columns. This pattern indicates that the text-guided selection mechanism is driven by localized semantic peaks rather than diffuse similarity. Importantly, Figure 2b visualizes the downstream effect of this semantic filtering on the raw interview audio (red segments): only the participant-response intervals temporally aligned with the

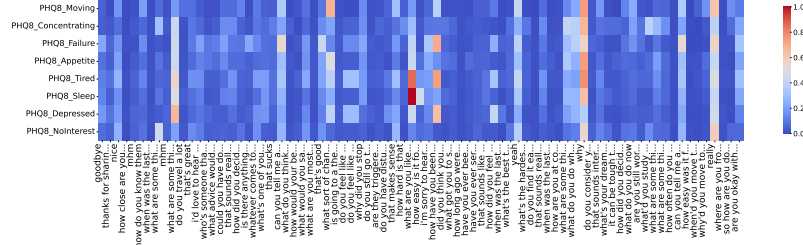
selected symptom-relevant prompts are retained, yielding a compact set of waveform fragments that reduces semantically irrelevant variability before MIL aggregation. Nevertheless, the heatmap also reveals a coverage limitation intrinsic to the DAIC-WOZ interview script: PHQ dimensions that are weakly or indirectly probed tend to show weaker or less isolated maxima, implying that for some symptoms the selector may rely on implicit cues or secondary matches. This bottleneck can propagate as selection noise into the MIL stage.

Then, the depression detection results shown in Figure 2c demonstrate that the proposed TCAMIL framework achieves the most consistent and balanced performance. In particular, TCAMIL attains the highest AUC (approx. 0.74), indicating superior discriminative capability across decision thresholds, together with the strongest F1-score and recall (both close to 0.75). This improvement reflects a substantially enhanced ability to correctly identify depressed subjects, which is critical in imbalanced clinical settings. Precision is also notably increased relative to the baselines, suggesting that the PHQ-conditioned semantic filtering stage effectively reduces false positives by restricting the MIL bags to clinically relevant segments. In contrast, classical models such as SVM and Random Forest exhibit competitive accuracy values; however, their lower F1 and recall scores reveal a bias toward the majority class, indicating that their accuracy is partially inflated by class imbalance. FAD3QN, which relies on randomly sampled segments without semantic selection or attention-based aggregation, shows weaker recall and overall instability, highlighting the limitations of random segment aggregation for capturing diagnostically salient evidence. The standard Attention-MIL model improves minority-class detection compared to classical baselines, confirming the advantage of learnable instance weighting under weak supervision; nevertheless, without symptom-guided filtering, it remains sensitive to semantically irrelevant variability present in long-form interviews. Overall, these results indicate that integrating PHQ-conditioned semantic selection with attention-based MIL produces a more clinically grounded and discriminative approach.

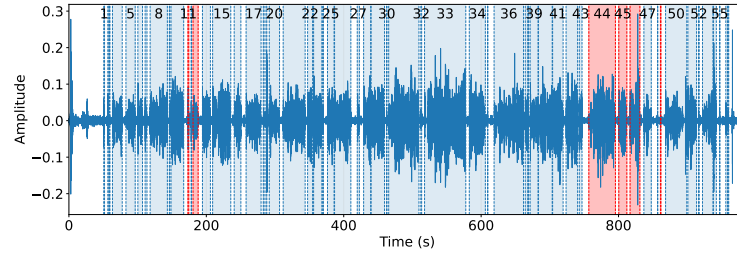
## 5 Conclusions

This work introduced a text-guided acoustic MIL framework, termed TCAMIL, for depression detection that integrates PHQ-8-conditioned semantic alignment prior to attention-based aggregation. By explicitly grounding instance selection in clinically structured symptom descriptors, the proposed approach reduces semantically irrelevant variability in long-form interviews and constructs symptom-aligned MIL bags for weakly supervised learning. Experimental results on DAIC-WOZ demonstrate that this semantic filtering strategy improves discriminative capacity, yielding superior recall, F1-score, and AUC compared to classical classifiers, random segment aggregation, and standard Attention-MIL without text conditioning. The analysis of the PHQ-8 latent structure further supports the stability of symptom-based anchors and the robustness of the framework against demographic confounds. Overall, the proposed method provides a clinically interpretable and structurally grounded alternative to purely data-driven aggregation strategies for audio-based depression detection. Future work will explore multi-instance retention per symptom dimension, multi-modal learning, coverage-aware semantic re-ranking, and joint optimization of semantic and acoustic modules to further improve recall while preserving clinical interpretability.

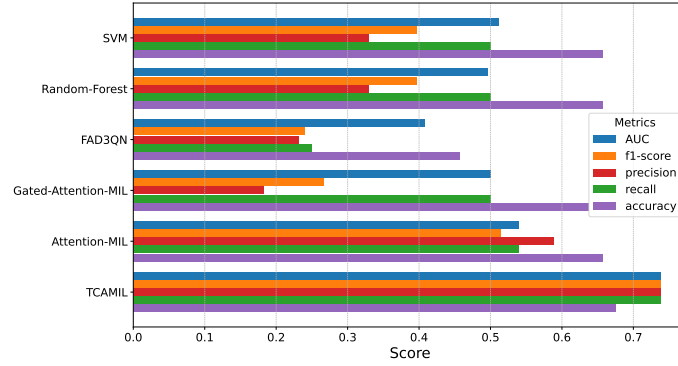
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(a) PHQ-conditioned semantic matching (Min-Max scaling).



(b) Full interview waveform with PHQ-aligned temporal segments.



(c) Comparative evaluation of depression detection models on DAIC-WOZ.

Fig. 2: TCAMIL results. (a) PHQ-aligned instance selection. (b) Audio segments retained for MIL aggregation. (c) Depression detection comparison.

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